

Joel Lawore

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EDUCATION

Tufts University

Bachelor of Science in Computer Science and Mathematics; Minor in Economics

May 2027

GPA: 3.5/4.0 | Dean's List, 5 semesters

PROFESSIONAL EXPERIENCE

Wabtec Corporation

LEAD IT Intern - Enterprise AI & Customer Service Transformation

Jun 2026 – Jul 2026

- Built a configurable Python framework using Salesforce APIs and a ground-truth benchmark of approximately 150 cases across 15 representatives, establishing the team's first repeatable process for measuring AI classification accuracy; selected for use in an upcoming proof of concept.
- Served as project manager and automation engineer for the winning intern technology challenge team, developing a Python/ML analytics pipeline to identify employees suitable for Admin By Request allowlisting in an environment where ~99% of employees remained on the higher-risk full admin-request workflow.
- Productionized and transitioned the solution to stakeholder owners by adapting its Python data/ML pipeline for GitLab CI/CD, containerizing SQL dependencies with Docker, configuring scheduled runner execution and secure API credential handling, and automating processed-data uploads to SQL for Power BI visualization.

Engineering Intern - Software Engineering

May 2025 – Aug 2025

- Built Python workflow automation synchronizing issue status, ownership, and workflow data between ClearQuest and GitLab to reduce duplicate-entry risk; presented the solution to senior leaders across multiple application teams and transitioned it to a full-time engineer for continued development.
- Prototyped a Docker-based local integration-testing framework with a user interface to simulate service-to-service inputs under production-like conditions; transitioned the working prototype to a full-time associate.
- Diagnosed and resolved 3 defects in C-based microservices deployed through Docker and integrated using Data Distribution Service (DDS).

Tufts University

Teaching Assistant, CS105: Programming Languages

Sep 2025 – Jan 2026

- Led office hours three times weekly and graded assignments for a course of 150+ students, explaining functional programming, type systems, and interpreter design.

SELECTED TECHNICAL PROJECTS

uScheme Interpreter Optimization (Standard ML)

- Implemented recursive free-variable analysis across the language AST and filtered closure environments to retain only referenced bindings, reducing interpreter runtime by 74% from 5.14s to 1.35s on the assignment benchmark.

Universal Machine Performance Optimization (C, Valgrind/Callgrind)

- Profiled and optimized an unfamiliar 32-bit virtual-machine implementation by replacing higher-overhead sequence/UInt representations with arrays, consolidating execution to improve locality, debugging stack exhaustion via heap allocation, and inspecting generated assembly to validate jump-table dispatch and register use.

LEADERSHIP

Tufts Daily Flame - Co-President: Lead 6–7 students across Men of Iron, events, and YouTube for a 100+ student fellowship; founded and grew Men of Iron from 3–4 active men to 10–15 weekly attendees.

Tufts Black Men's Group - Co-President: Built a 7-member leadership structure and governance framework adopted by the Africana Center.

SKILLS & INTERESTS

Technical: Python, C, C++, Standard ML, Java, Git, GitLab CI/CD, Docker, Linux, REST/Salesforce APIs, Valgrind/Callgrind, Power BI

Languages: Spanish (conversational) | **Interests:** Arsenal FC, music & DJing, philosophy, coffee, financial markets